Antonio Aguirre

Santa Cruz, CA | +1 (831) 515-9264 | antonio.agprz@gmail.com | jaguir26@ucsc.edu
antonio-aguirre.com | github.com/antonioapdl | linkedin.com/in/antonioapdl

Summary

Ph.D. candidate in Statistics with experience in Bayesian modeling, time series forecasting, uncertainty quantification, and scalable inference. Industry background across finance, sensor analytics, and research computing, with strong Python/R/Linux/Git workflows and a focus on reproducible analysis, diagnostics, and decision-facing reporting.

Education

Ph.D. in Statistics, University of California, Santa Cruz — Expected Aug. 2026
Research: Bayesian time series, quantile modeling, forecast correction and synthesis, scalable variational Bayes inference. GPA: 3.82/4.00.

M.Sc. in Economics, Instituto Tecnológico Autónomo de México (ITAM) — 2018–2020
Mexico City, Mexico

B.Sc. in Applied Mathematics, Instituto Tecnológico Autónomo de México (ITAM) — 2014–2018
Mexico City, Mexico

Industry & Professional Experience

Delos Financial Technologies, San Rafael, CA (Remote) — May 2025–Sep. 2025
Quantitative Researcher (Contract) — *Python, Git, Bash, C++, AWS, Stan*

- Built evaluation notebooks for goodness-of-fit, residual diagnostics, and cohort-level reporting to standardize model comparison.
- Implemented and validated baseline and extended model variants, refactoring interfaces for consistent parameterization and downstream plotting.
- Automated backtests and experiment runs on AWS with versioned outputs to improve reproducibility and reviewability.
- Investigated numerical stability issues and proposed mitigation strategies for forecasting workflows.

NeatLeaf Inc., Scotts Valley, CA — May 2021–Sep. 2022
Data Analyst — *Python, SQL, Git, Bash, Linux, C++, DBeaver*

- Built data pipelines for real-time greenhouse microclimate telemetry and downstream analytics.
- Developed spatiotemporal statistical models for air anomaly detection and delivered actionable diagnostics.
- Analyzed robot performance and sensor calibration to improve data quality and system reliability.

Banco de México (BANXICO), Mexico City, Mexico — Aug. 2018–Dec. 2019
Data Analyst — *Python, R, SQL, Git*

- Built data pipelines for banknote image datasets used in operational monitoring and model development.
- Developed statistical and machine learning models for anomaly classification and diagnostic evaluation.
- Built forecasting prototypes for financial volumes and tools for model selection and performance tracking.

UCSC Statistics Department, Santa Cruz, CA — Feb. 2024–Present
Computer Systems Coordinator — *Linux, Bash, Python/R, Networking, User Support*

- Administer Linux servers supporting faculty and graduate research, including upgrades, storage, software stacks, and incident resolution.
- Built automation scripts and internal documentation for onboarding, resource usage, and reproducible compute workflows.
- Coordinated with central IT on infrastructure planning and served as the primary technical contact for departmental computing support.

Selected Publications & Presentations

Aguirre, A., & Lobato, I. N.
Evidence of non-fundamentalness in OECD capital stocks.

Empirical Economics 67, 761–772 (2024). doi:10.1007/s00181-024-02564-5

Aguirre, A., Sansó, B., & Prado, R.
Bayesian Quantile-Based Correction and Synthesis of River Flow Forecasts.
Manuscript in revision.

ISBA World Meeting 2026, Nagoya, Japan
Poster presentation accepted: "Bayesian Quantile-Based Correction and Synthesis of River Flow Forecasts."
Scheduled poster session: July 1, 2026.

LACSC-TIES-EnviBayes-EnvrASA 2026, Mexico City, Mexico
Invited-session abstract submitted as a 2026 EnviBayes Student Paper Competition winner.
EnviBayes competition-winners invited session.

Awards, Fellowships & Competitive Funding

- **ISBA 2026 Junior Researcher Travel Award**, International Society for Bayesian Analysis / National Science Foundation, 2026 ($1,000).
- **EnviBayes Student Paper Competition Winner**, International Society for Bayesian Analysis, 2026 ($250 award).
- **UCSC Statistics Department Conference Travel Award**, 2026 ($1,000).
- **UCSC Summer Research Fellowship**, UCSC Statistics, 2024.
- **UC Regents Fellowship**, UCSC Statistics, 2021.
- **TLC Graduate Pedagogy Fellowship Certificate**, Center for Innovations in Teaching and Learning, 2023–2024.
- **Baillères Fellowship**, ITAM B.Sc., 2014–2018.
- **Baillères Fellowship**, ITAM M.Sc., 2018–2020.

Teaching & Mentoring

University of California, Santa Cruz — 2021–Present
Teaching Assistant / Graduate Student Instructor — *Statistics and Probability*

- Supported or taught courses in probability theory, introductory statistics, Bayesian inference, and data visualization.
- Graduate Student Instructor for *Data Visualization (STAT 80B)*, Spring 2025.
- Designed worksheets, grading rubrics, and discussion materials; supported instruction across lower- and upper-division courses.

ASA DataFest, Fresno, CA — Apr. 2023
Mentor — *R, Python*

- Mentored student teams on modeling and visualization and provided structured technical feedback.

Professional Development

- **Summer Institute in Statistical Genetics (SISG)**, University of Washington / NSF, 2022.
- **Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)**, University of Washington / NSF, 2022.
- **Fundamentals of Causal Inference (with R)**, ASA / Genentech, 2023.
- **Professional Communication Certificate Program (PCCP)**, UCSC Extension, 2021.

Technical Skills

Programming	Python, R, SQL, C/C++, Bash, Git, LaTeX
Methods	Bayesian inference, time series forecasting, quantile modeling, model and variable selection, variational inference, diagnostics, calibration
Tools	AWS (EC2, EMR, S3), Linux, Stan, Tableau
Languages	English, Spanish, German